

Altium ***Designer***

Module 22: Signal Integrity

Module 22: Signal Integrity

22.1 Signal Integrity.....	22-1
22.2 Checking Signal Integrity on an FPGA design.....	22-3
22.2.1 Setting Up.....	22-3
22.2.2 Importing IBIS Models	22-3
22.2.3 Initial IO Standard, Drive and Slew settings.....	22-4
22.2.4 Simulating the Reflecting Characteristics.....	22-6
22.2.5 Checking for Crosstalk	22-10
22.2.6 Finding coupled Nets.....	22-11
22.2.7 Performing Crosstalk Analysis	22-13
22.2.8 Summing Up.....	22-14

Software, documentation and related materials:

Copyright © 2008 Altium Limited.

All rights reserved. You are permitted to print this document provided that (1) the use of such is for personal use only and will not be copied or posted on any network computer or broadcast in any media, and (2) no modifications of the document is made. Unauthorized duplication, in whole or part, of this document by any means, mechanical or electronic, including translation into another language, except for brief excerpts in published reviews, is prohibited without the express written permission of Altium Limited. Unauthorized duplication of this work may also be prohibited by local statute. Violators may be subject to both criminal and civil penalties, including fines and/or imprisonment. Altium, Altium Designer, Board Insight, Design Explorer, DXP, LiveDesign, NanoBoard, NanoTalk, P-CAD, SimCode, Situs, TASKING, and Topological Autorouting and their respective logos are trademarks or registered trademarks of Altium Limited or its subsidiaries. All other registered or unregistered trademarks referenced herein are the property of their respective owners and no trademark rights to the same are claimed.

Module Seq = 22

22.1 Signal Integrity

To help designers cope with the added complexity of transmission line analysis on a PCB, EDA vendors are starting to include signal integrity analysis software as part of the standard board design toolset.

- Unlike circuit simulation, signal integrity analysis does not concern itself with the functional operation of the circuit – components are only modeled in terms of the I/O characteristics of their pins, without regard to the component's function.
- Connections between component pins are modeled using transmission line techniques that factor in the length of the trace, the characteristic impedance of the trace at the stimulus frequency, and the termination characteristics at each end of the connection.
- Traditionally, signal integrity tools are designed to work with a fully routed board. While this gives accurate results because each individual trace length is known, it does mean that the analysis is performed quite late in the design cycle. One of the most common signal integrity issues that arises is signal degradation resulting from reflection caused by mismatched impedances of the pins at either end of a connection. The solution to this problem usually involves the addition of a termination resistor or R/C network to match the termination impedances and minimize reflections – a task which requires modifying the source schematic. Altium Designer supports schematic-level signal integrity analysis, to help preempt potential impedance miss-matches before the board layout stage.

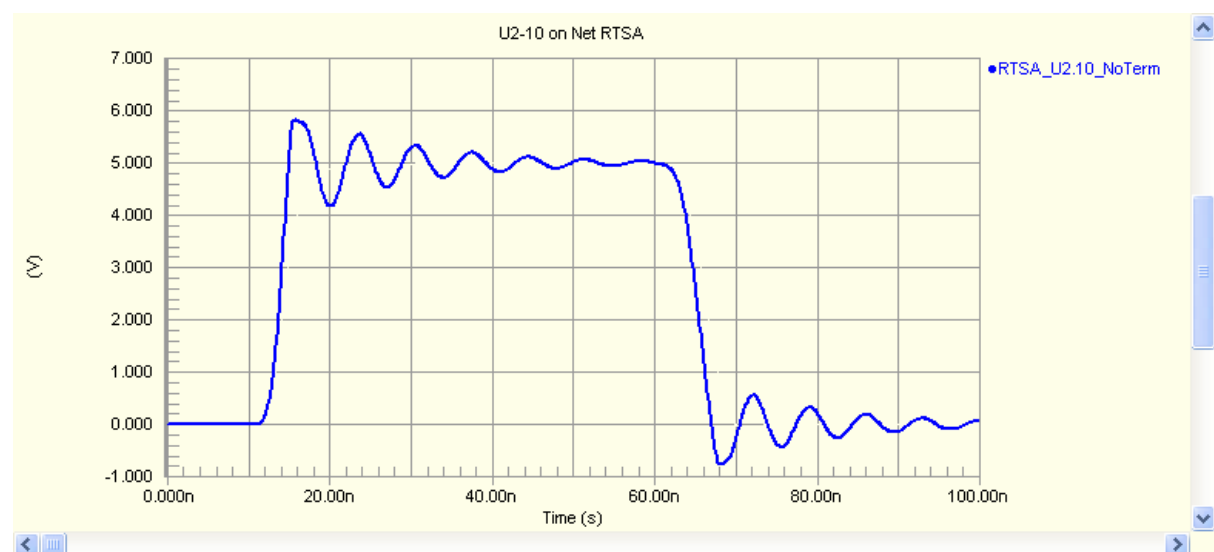


Figure 1. Output from a reflection simulation for a problem net on a board.

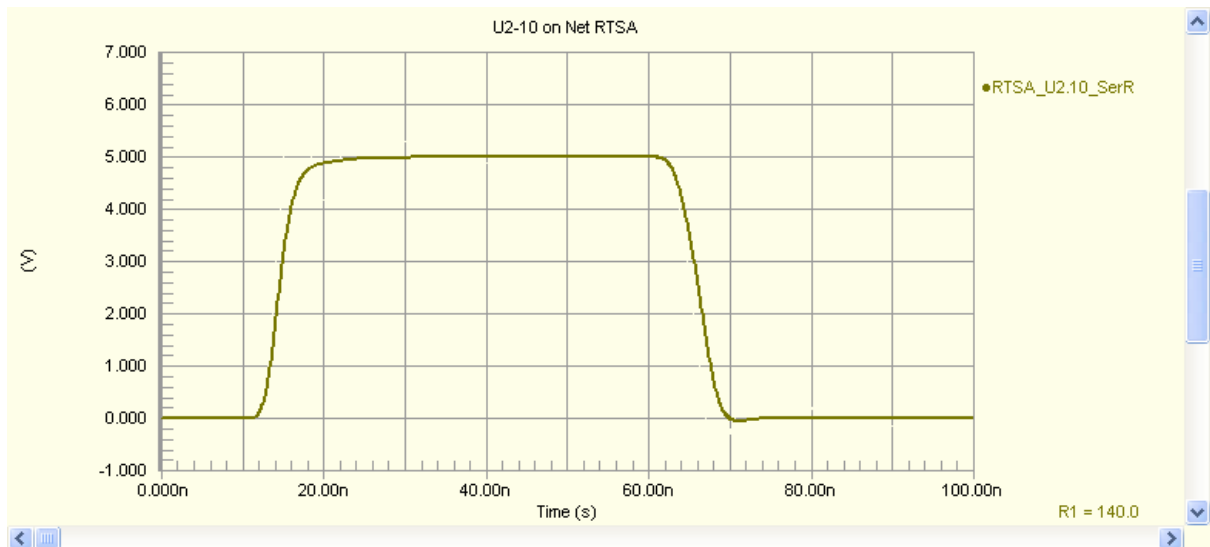


Figure 2. Same net after the insertion of an appropriate series termination resistor.

- Using signal integrity simulation to uncover problems in a completed board design before any prototyping is undertaken can reduce the number of prototype iterations needed to complete a project. However, the addition of new components late in the board design stage can be a major problem, particularly on dense boards. The rework involved can be time consuming, partially negating the time saved in the prototype/test phase.
- For this module we'll run through an exercise that takes you through the steps of setting up, finding and fixing signal integrity problems on a design using an FPGA, SDRAM and a discrete processor.

22.2 Checking Signal Integrity on an FPGA design

This tutorial relates to the following example project:

```
\Program Files\Altium Designer Summer 08\Examples\Signal Integrity\NBP-28\NBP-28.PrjPcb
```

This example is based on the daughterboard, NBP28. This daughterboard includes a Xilinx Spartan 3, a Sharp LH79520 incorporating an ARM 7 processor, SRAM and Flash RAM.

In this example we are going to answer the question *how hard can I drive the signals D[31..0] before ringing and crosstalk will prevent correct operation? Or alternatively, what is the optimum slew and drive settings to use for FPGA pins driving signals D[31..0]?*

22.2.1 Setting Up

Before we do any analysis we need to make sure that the components connected to the signals of interest all have the correct models set up.

These components are listed below.

- U1 FPGA - Xilinx XC3S1000-4FG456C
- U5 PROCESSOR - Sharp LH79520
- U7, U9 SRAM - Micron MT48LC16M16A2FG-75
- U8 FLASH RAM - AMD AM29LV640DU90RWHI

U1 is the FPGA. Suitable pin models for all FPGAs supported by Altium Designer are included with the software. The actual pin model that is automatically chosen depends on the final program state of the device, and the settings for pin IO standard, slew rate and drive strength. Part of this example is to determine the optimum settings for these.

22.2.2 Importing IBIS Models

You will need to import the IBIS models for the other ICs listed above in order to have the pin models added to the SI pin model libraries in your installation of Altium Designer. The required IBIS models have been downloaded from the vendor websites and can be found in the `\ibis models` folder in the example project directory. The IBIS model is imported by editing the schematic component.

Locate U5 on the schematics, and:

1. Double click to open the **Component Properties** dialog.
2. Edit the existing Signal Integrity model, the **Signal Integrity Model** dialog will open.
3. Click the Import IBIS button.
4. Locate and select the IBIS model file for the device.
5. Select the correct component if a choice is given in the upper region of the **IBIS Converter** dialog.
6. Click OK to exit the dialogs. You will be informed that the model data has been written to the libraries, and that the model has been assigned. A report file will open, listing which pin model has been assigned to each device pin.

7. Now repeat this process for U7, U8 and U9. Note that we are assuming that the IBIS models from the manufacturer accurately model the pin characteristics. As with any simulation, the models used are extremely important to accuracy. A simulation is only as accurate as the models used!

Note: For the SRAM, there are 3 possible IBIS model files. For this example, we will use `y16a.ibs`.

- Another essential requirement for successful SI simulation is to set up supply nets design rules for each of the supply nets (**Design » Rules**), so that the Signal Integrity Analyzer will handle them correctly. It is particularly important to set up supply net rules for the plane layers. These design rules have already been configured in the example board.
- The layer stack for the PCB must also be defined correctly. The Signal Integrity Analyzer requires continuous power planes. Split planes are not supported, so the net that is assigned to the plane is used. If they are not present, they are assumed, so it is far better to add them and set them up appropriately. The thickness for all layers, cores and prepreg must also be set correctly for the board. These properties, as well as dielectric values, can be defined in the *Layer Stack Manager* dialog (**Design » Layer Stack Manager**). All such settings have been defined already for the example board.

22.2.3 Initial IO Standard, Drive and Slew settings

The electrical properties for pins of the physical FPGA device are defined in the FPGA Signal Manager dialog Figure 3. Access this dialog from the schematic or PCB document using the Tools » FPGA Signal Manager menu command.

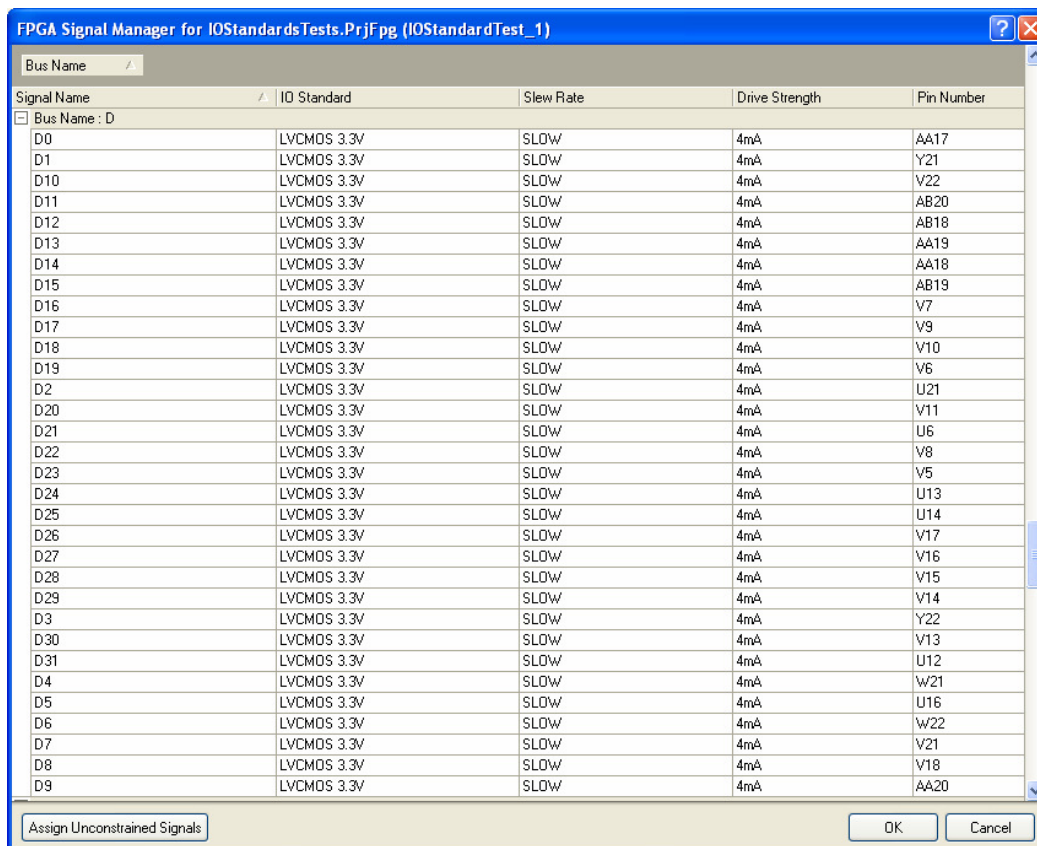


Figure 3. Controlling FPGA pin electrical characteristics using the FPGA Signal Manager.

- [illegible]

22.2.4 Simulating the Reflecting Characteristics

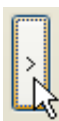
Now that things are setup we can simulate the reflection characteristics of these signals.

Firstly, to analyze the reflection characteristics:

1. Open one of the project documents (either a schematic or the PCB)
2. Run Tools » Signal Integrity
3. Click the **Continue** button when the **Errors or Warnings Found** dialog pops up. This dialog opens because there are components that the Signal Integrity Analyzer cannot correctly guess a suitable model for, in this case it is a number of test points – we can ignore these as they are not part of the nets we are interested in.

Note: The design will be analyzed and the **Signal Integrity** panel will open, listing all the nets in the design. What has happened is that a fast analysis has been performed on *all* nets in the design, called a screening analysis, with the results being listed down the left side of the **Signal Integrity** panel. Screening is used to quickly identify potential problem nets, which can then be analyzed in more detail. To analyze a net in more detail it is taken over to the right hand side of the panel, where a reflection or cross talk analysis can be performed. Note that a number of nets have a status of Not Analyzed, typically this is because these nets include a connector pin.

As part of the screening analysis results, you will see that failed nets have red backgrounds to their cells. For Overshoot and Undershoot conditions, the Signal Integrity engine has built-in thresholds that it can test against. These are configured in the *Set Screening Analysis Tolerances* dialog. Access this dialog by clicking on the **Menu** button and choosing the **Set Tolerances** command.



4. Locate the net D0, and click the Take Over button at the top of the panel to add D0 to the right hand section of the panel.

5. Make sure that the direction of the FPGA pin on this net is set to Bi/Out, and that all other pins are set to Bi/In. Note that the FPGA is component U1. To change this, right click in the pins on each pin that needs to be changed, and choose "Toggle In/Out" to change the status. The reason for this is that we want the FPGA pin to be driving the net for the signal integrity analysis.

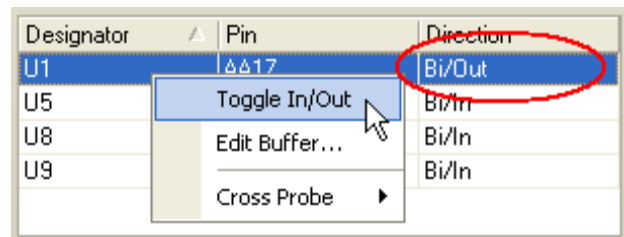


Figure 5. Set the FPGA pin to Bi/Out, and the rest of the pins in the net to Bi/In.

6. Click the **Reflections** button at the bottom of the panel to run a Reflection Analysis on the net D0. The resulting waveform will appear, as shown in Figure 6.

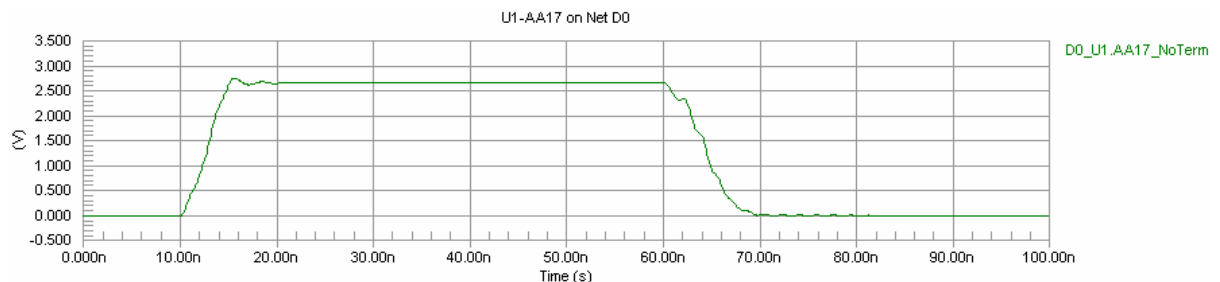


Figure 6. Waveform for D0 at the FPGA pin, with Slew = SLOW and Drive = 4mA.

7. Locate the signal D0 in the *FPGA Signal Manager* dialog and set the Slew Rate to the FAST option. Click **OK** and then **Execute** the subsequent engineering change order that is generated. The associated constraint file will be updated with the new setting and opened as the active document. Save and close this file.
8. Display the Signal Integrity panel again (you can click the **Signal Integrity** button down the bottom of the workspace if you closed the panel), and click the **Reanalyze Design** button.
9. Ensure that the FPGA pin is still set to be the output pin on the net (Bi/Out). Then run a reflection analysis on D0 again and note that there is now some ringing on this signal (Figure 7).

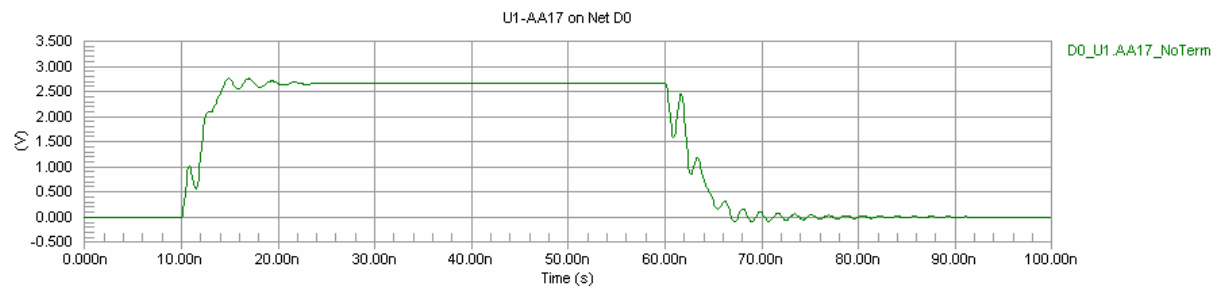


Figure 7. Waveform for D0 at the FPGA pin, with Slew = FAST and Drive = 4mA.

So just how much ringing can we allow on these signals in order to avoid glitches on the input pins? Consulting the datasheets for components U5, U7 and U9 reveals the following characteristics:

- Maximum VIL = 0.8
- Min VIH = 2.0

In the NBP-28.sdf waveform document you will find Charts for a range of possible Slew and Drive Strength settings. There is also a chart named "Comparison" with all the different waveforms combined for comparison. To scroll through the different waves, select a wave name and use the mouse scroll wheel or up down arrows.

Note: The naming of the waveform in the comparison follows the following format: <signal>_<comp designator>.<pin>_<slew>_<drive> For example D0_U1.AA17_F_6mA is the waveform for signal D0, component U1, pin AA17, Fast slew rate, 6mA drive strength.

The comparison graph was generated using the **File » Export » Chart** command to create a CSV file for each of the charts. The wave names were changed inside this file to follow the above convention, and then the CSV files were all imported into a new chart.

From browsing these waveforms we can see that most of the FAST slew rate waveforms are not acceptable as they have excessive ringing present on the signals. With the slew rate set to SLOW, 6mA drive strength gives the cleanest looking waveform (Figure 8).

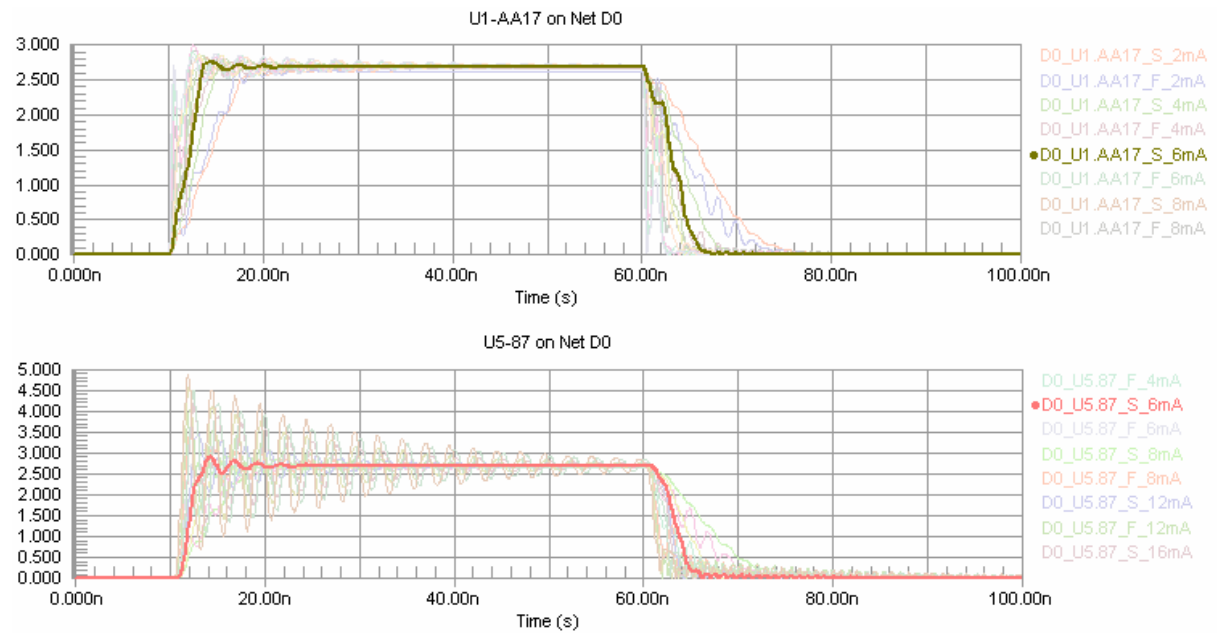


Figure 8. Waveform for D0 at the FPGA and Processor pins, with Slew = SLOW and Drive = 6mA.

A drive strength of 8mA would probably be acceptable although there is some ringing (Figure 9).

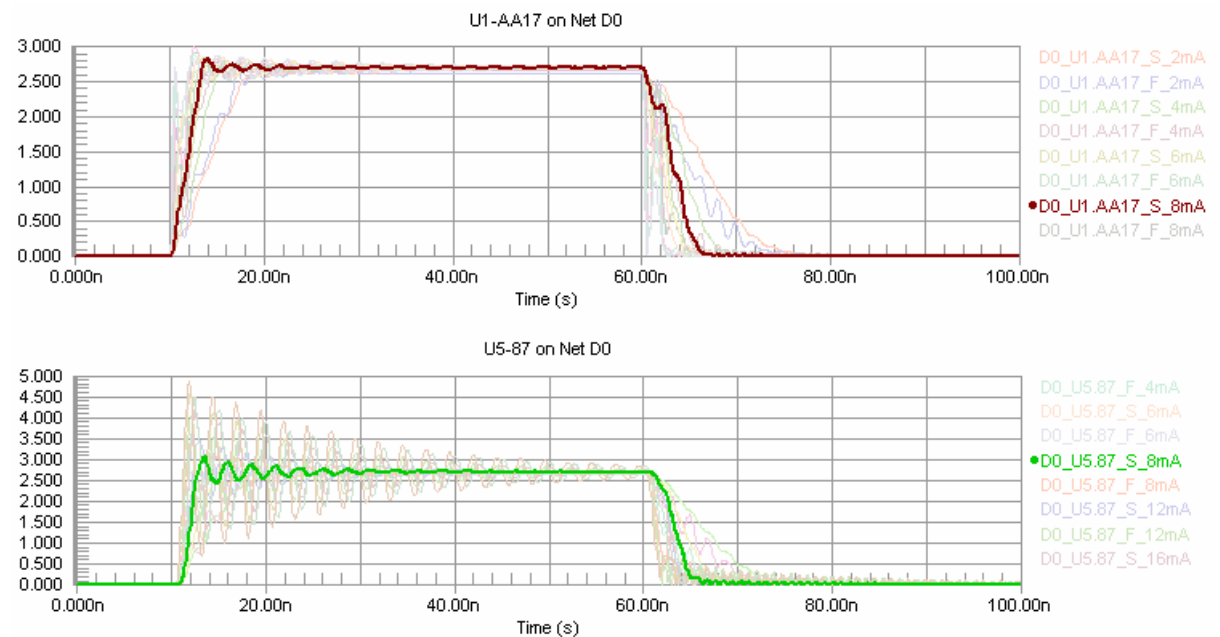


Figure 9. Waveform for D0 at the FPGA and Processor pins, with Slew = SLOW and Drive = 8mA.

Any drive strengths above this generate large amounts of ringing, particularly on U5, pin 87 (Figure 10).

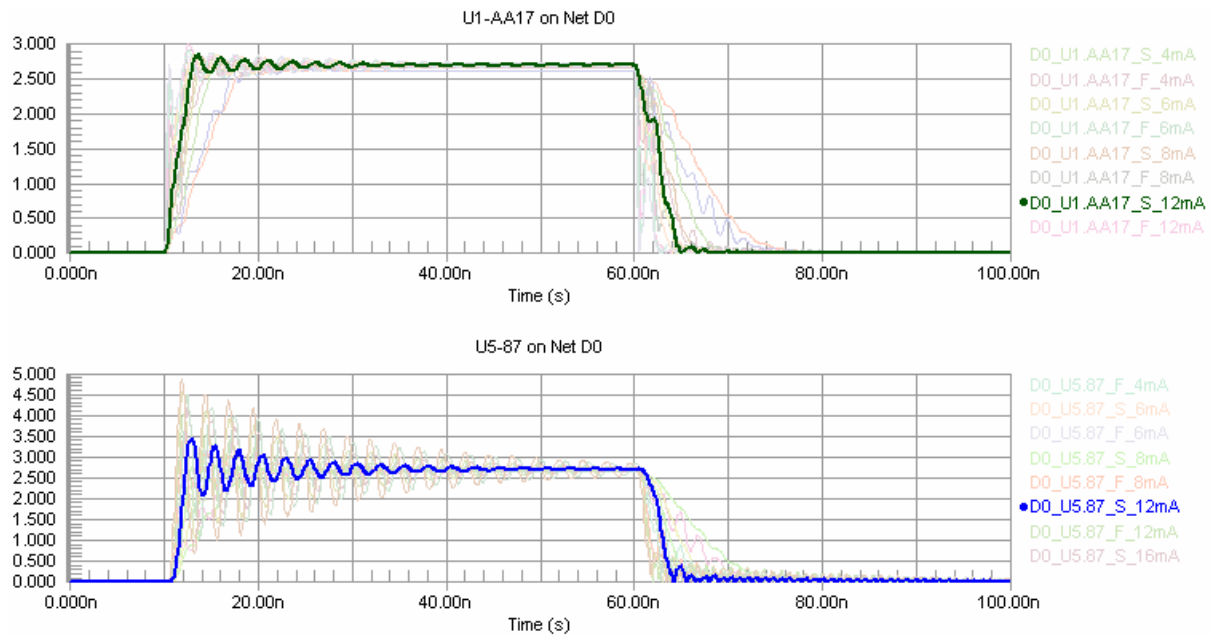


Figure 10. Waveform for D0 at the FPGA and Processor pins, with Slew = SLOW and Drive = 12mA.

You can see from the waveforms that the slew rate and drive strengths can have a huge effect on how clean the signals are. By modifying the slew and drive characteristics we can select values that provide a suitably fast and clean response.

Given the above findings we would select SLOW slew rate and 6mA drive strength for the FPGA pins. We would definitely not want to drive it any harder than 8mA or use the FAST slew rate setting without adding terminations on the PCB.

Selecting SLOW Slew and 6mA drive strength for all the FPGA signals D[31..0] in the signal manager we can now verify that this gives satisfactory waveforms for all these signals.

1. Make these changes in the *FPGA Signal Manager* dialog.
2. Re-analyze the design with the Signal Integrity tool.
3. Take across signal D[31..0]
4. Remember to check the i/o direction of the pins, make sure that for each signal the FPGA pins are set to Bi/Out and all other pins are Bi/In
5. Click the **Reflections** button. This generates a chart of reflection waveforms for each of the selected signals.

Examining the generated waveforms shows that 6mA drive strength gives acceptable waveforms for all these signals.

If we wanted to drive the pins harder than this then it would be necessary to place terminations on the PCB. The termination advisor is a useful tool for deciding upon a type of termination to use, and it can be used to sweep values of termination components across a range.

Using the termination advisor:

1. Using the *FPGA Signal Manager* dialog, set the drive current for D0 to 24mA.
2. Re-analyze the design with the Signal Integrity tool
3. Take across signal D0.
4. Make sure that the FPGA pin AA17 is set to Bi/Out, and all other pins are set to input.
5. Enable all the different termination types.
6. Un-check the **Perform Sweep** checkbox.

7. Click the **Reflections** button, and waves will be generated for each termination type.
8. From the results we can see that a serial resistor looks like it will fix the ringing.
9. In the Signal Integrity panel, disable all terminations except for Serial Resistor.
10. Check the **Perform Sweep** option.
11. Click the **Reflections** button.
12. Click on the name of the first wave in the legend, and use the mouse wheel to scroll through the waves for different termination resistance values. It would appear that a value around 47 Ohms would work well, so let's perform a reflection simulation with this value.
13. Uncheck the **Perform Sweep** option.
14. Change the value of the Serial Resistor to 47.
15. Run the reflection analysis.

As you can see, this can be a useful tool for selecting terminations in order to solve SI issues. The values used for each sweep are shown in the waveform hint for each wave. In the above example we can see that a value of 47 ohms for the serial resistor should provide a good termination if we needed to use the high drive current option for the FPGA output pin. In `NBP-28.sdf` there is a chart named Comparison2. This contains a comparison of the 24mA drive current terminated with a 47 Ohm serial resistor with the 6mA drive current waveforms (Figure 11).

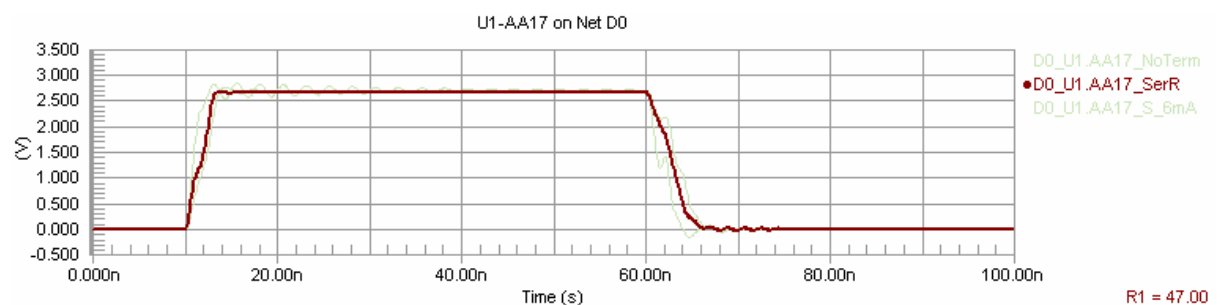


Figure 11. Waveform for D0 at the FPGA pin, with Slew = SLOW, Drive = 24mA and 47Ohm Series Resistor.

There is not much difference between these signals, so unless we needed the higher drive current we are best off sticking to the 6mA and avoiding the need to add termination resistors.

22.2.5 Checking for Crosstalk

Finally we will look at how to examine the crosstalk between signals. Given that we were able to choose drive and slew settings that minimized ringing on the signals, crosstalk problems should have already been minimized.

There are essentially two stages in performing a crosstalk analysis. First, we must identify the nets for which crosstalk may pose a problem. Then we must take over the nets which we want to more closely examine, and run a detailed crosstalk analysis.

22.2.6 Finding coupled Nets

The Signal Integrity panel includes a feature that enables you to quickly identify which nets are considered to be coupled, to the selected net(s) you choose in the screening analysis results area of the panel. This feature – Find Coupled Nets – is ideal for working out which nets may be susceptible to crosstalk. It essentially analyzes the PCB and identifies traces that run parallel to each other according to coupling options defined on the Configuration tab of the Signal Integrity Preferences dialog (Figure 12). Access this dialog by clicking on the Menu button in the panel, and choosing Preferences from the resulting menu.

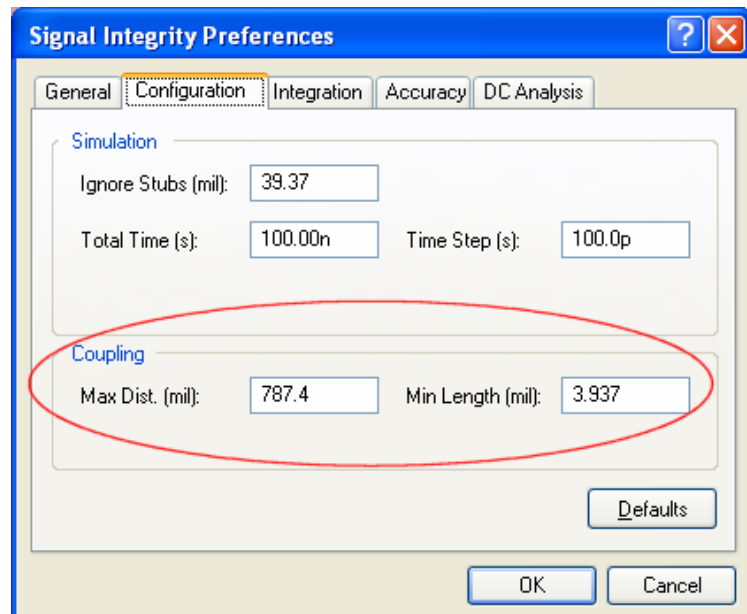


Figure 12. Set criteria to identify coupled nets.

Two options constitute the coupling criteria:

- **Max Dist.** – this option defines the proximity for neighboring nets. Only nets within the specified distance of the selected base net will be considered for coupling purposes.
- **Min Length** – this option specifies the minimum length that traces of nets must run parallel to that of the selected base net, in order to be considered coupled.

As a general rule, the closer the traces the more likely crosstalk is to occur. Similarly, the longer the parallel sections of trace, the more crosstalk will occur.

Note: When running a crosstalk analysis, the typical number of nets taken across would be three – a net and its two immediate neighbors.

To illustrate the power of this feature, let us find all nets coupled to the net D5 (our base net). The default setting for the Max Dist is a little high and would easily find all nets in the D[31..0] bus, as well as many others. Instead, we will vary Max Dist – using 20mil and 10mil values – and observe the resulting nets that are deemed to be coupled. In addition we will set the Min Length to be 50mil (i.e. any nets not having traces running parallel to D5 for at least this distance will not be considered coupled).

To find coupled nets with **Max Dist** = 20mil, **Min Length** = 50mil:

1. Change the **Max Dist** coupling option to 20mil
2. Change the **Min Length** coupling option to 50mil
3. Select net D5 in the screening analysis region of the **Signal Integrity** panel
4. Right-click and choose **Find Coupled Nets**
5. Observe that the following data nets – D1, D4, D6 and D7 – are all found to be coupled to D5 and are selected ready to take across for a more detailed crosstalk analysis (**Error! Reference source not found.**). Net A17 is also found to be coupled, but as we are interested only in the data lines, this should be deselected prior to taking the nets across.

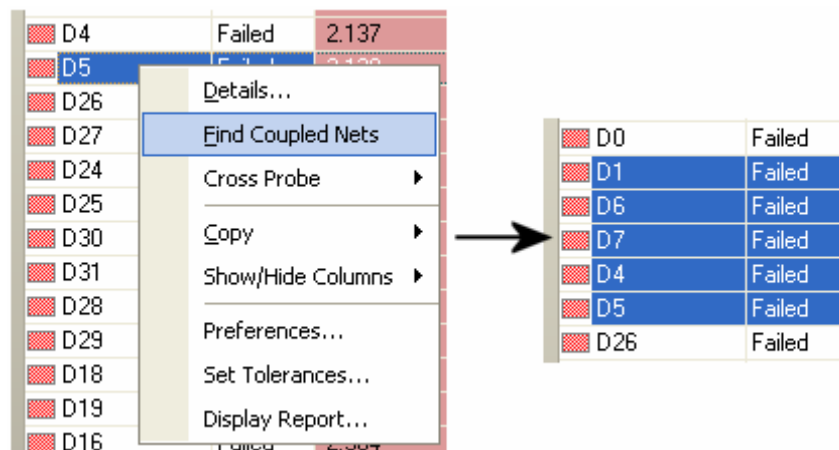


Figure 13. Nets coupled to D5 (Max Dist = 20mil, Min Length = 50mil).

To find coupled nets with **Max Dist** = 10mil, **Min Length** = 50mil:

1. Change the **Max Dist** coupling option to 10mil
2. Change the **Min Length** coupling option to 50mil
3. Select net D5 in the screening analysis region of the **Signal Integrity** panel
4. Right-click and choose **Find Coupled Nets**
5. Observe that the following data nets – D6 and D7 – are all found to be coupled to D5 and are selected ready to take across for a more detailed crosstalk analysis (**Error! Reference source not found.**).

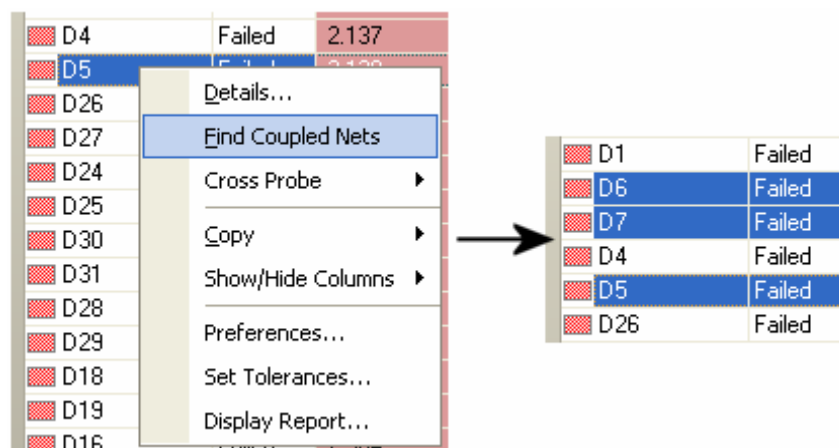


Figure 14. Nets coupled to D5 (Max Dist = 10mil, Min Length = 50mil).

22.2.7 Performing Crosstalk Analysis

In the previous section, we used the Find Coupled Nets feature to identify nets coupled to the net D5, and that may suffer from the effects of crosstalk. We will now perform crosstalk analysis to assess the level of crosstalk involved when the net D5 is set to Aggressor and is driven at different signal strengths. Note that although several coupled nets were found by the Find Couple Nets feature, we will concentrate on just examining D5 and D6, since these two nets run parallel to each other for much of their length.

1. Set the drive current of D5 to 6mA.
2. Analyze the design using the Signal Integrity tool.
3. Take over signals D5 and D6.
4. Make sure that all the FPGA pin directions are set to Bi/Out.
5. Right Click on D5 in the list of nets and select "Set Aggressor". This means that during crosstalk analysis we will be driving this net and simulating its effect on the other "victim" nets.
6. Click the **Crosstalk** button to run a crosstalk analysis.
7. In the generated chart we can see that up to about 125mV is induced in D6. (see chart "Crosstalk Analysis 6mA"). This maximum occurs at pin 79 of component U5 (Figure 15). As anticipated it should not be enough to cause any problems.

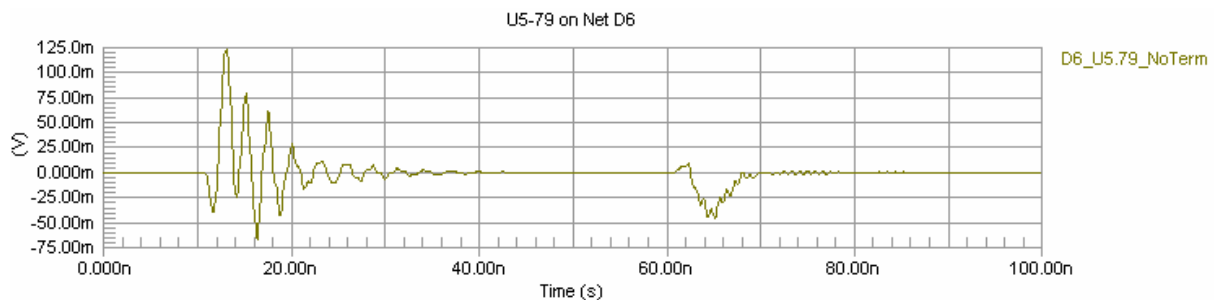


Figure 15. Crosstalk induced in D6 by D5 (D5 driven at 6mA – max crosstalk on pin 79 of U5).

If we repeat the above steps but with 24mA drive strength for D5 we see that the simulation waveforms show up to about 500mV is induced in the coupled nets (see chart "Crosstalk Analysis 24mA"). This maximum is again found on pin 79 of component U5 (Figure 16). This is quite significant and would certainly be likely to cause problems on the signal D6.

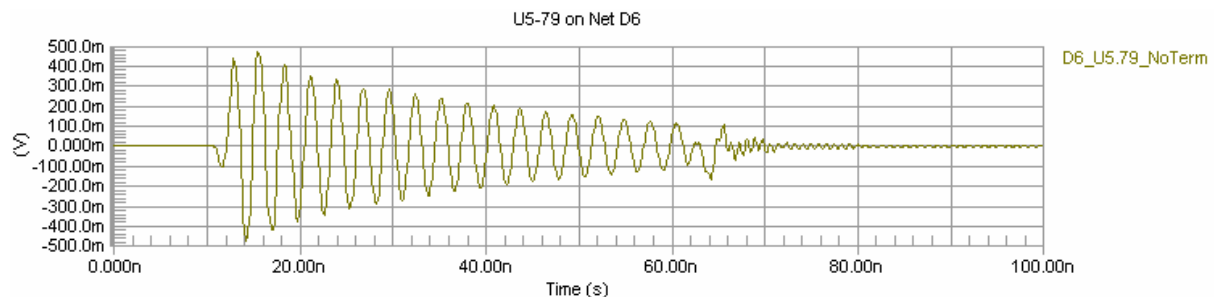


Figure 16. Crosstalk induced in D6 by D5 (D5 driven at 24mA – max crosstalk on pin 79 of U5).

You can experiment further with other slew and drive settings in the *FPGA Signal Manager* dialog and see the effect on crosstalk.

22.2.8 Summing Up

In conclusion we have used the Signal Integrity tool to find the optimum drive and slew settings for the FPGA pins D[31..0]. In this case we chose a SLOW slew rate and 6mA drive strength. We have also used the termination advisor to choose appropriate terminations if the higher drive current is required. We have also looked at how to assess crosstalk between signals.

For detailed information on the **Signal Integrity** panel, press **F1** while the panel has focus. Use the available link in the upper region of the **Knowledge Center** panel to access the relevant topic for this panel, within the *Altium Designer Panels Reference*.

For conceptual information regarding what is required to address possible signal integrity issues that may arise in a design, refer to the article *Putting Signal Integrity in its Place*.

For further tutorial-based information on Signal Integrity – including pre- and post-layout analysis – refer to the *Performing Signal Integrity Analyses* tutorial.

For further information on working with generated waveforms, refer to the application note *Working with the Sim Data Editor*.